

Group Report

[NXC5015: Structural Characterization]

IR Spectroscopy and Thermal Analysis



Group number	M5
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1. Determination of the samples

Before the formal analysis of each samples, the infrared spectroscopy and Thermogravimetric analysis spectrum were plot for determination of each samples.

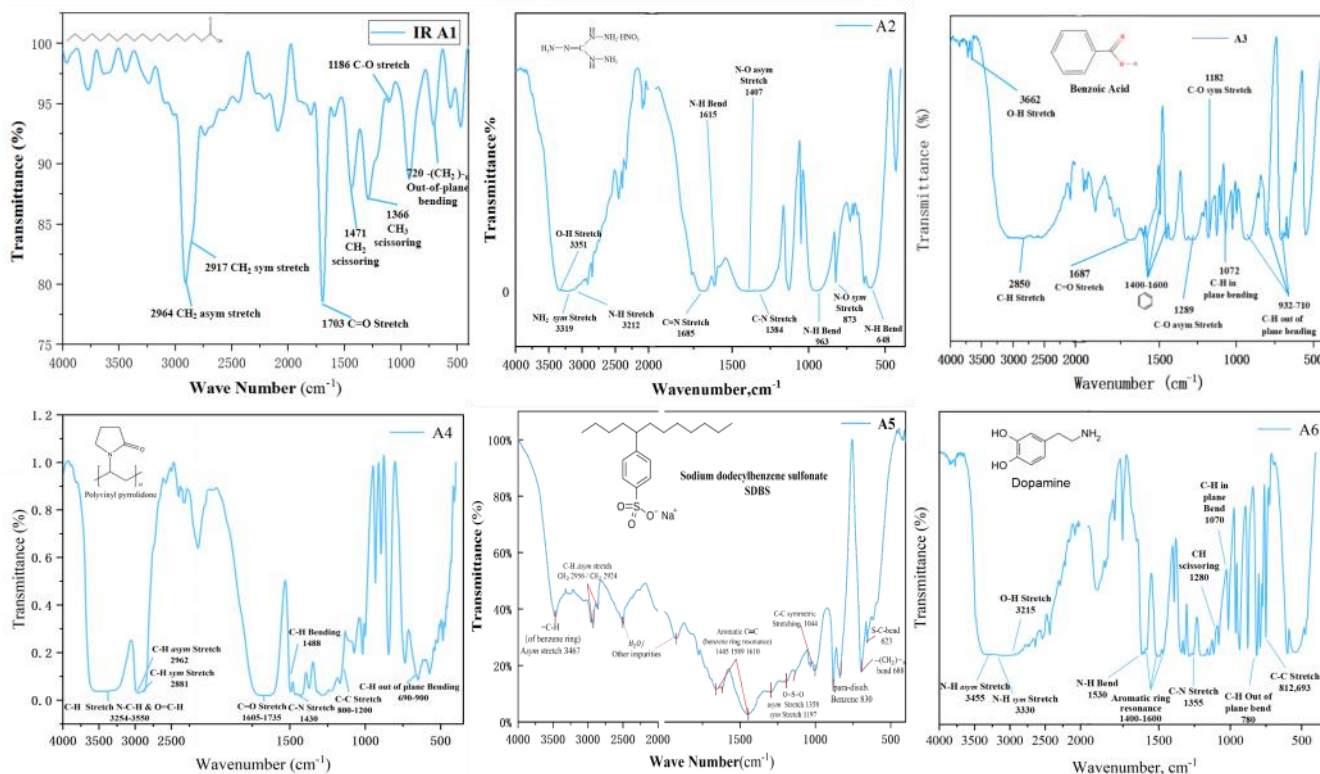


Figure 1. The infrared spectroscopy of the samples

Infrared spectroscopy was used as the primary means of analysis, and combined with data from thermogravimetric analysis, the six substances were eventually identified as results shown in Table 1.

Table 1. The determination of the material

Sample	Material
A1	Stearic acid
A2	TAGN
A3	Benzoic acid
A4	Polyvinyl pyrrolidone
A5	SDBS
A6	Dopamine

The steps for judgement can be summarize as follow:

- First, we have searched for the molecular structure, standard infrared spectroscopy and Thermogravimetric analysis spectrum for each undetermined samples. And learned about the basic information of the samples, including their functional groups and decomposition temperature etc.

- Second, the infrared spectroscopy and Thermogravimetric analysis spectrum are plot based on the data collected in the experiment by Origin and were compared to the standard spectrum.

- Follow step 1 and 2, most of the samples are determined. For two materials which have similar functional groups and decomposition temperature such as TAGN and Dopamine, the molecular structure of

these two samples were compared carefully, and the peaks caused by the benzene ring resonance of Dopamine was the most important factors to make the final judgement.

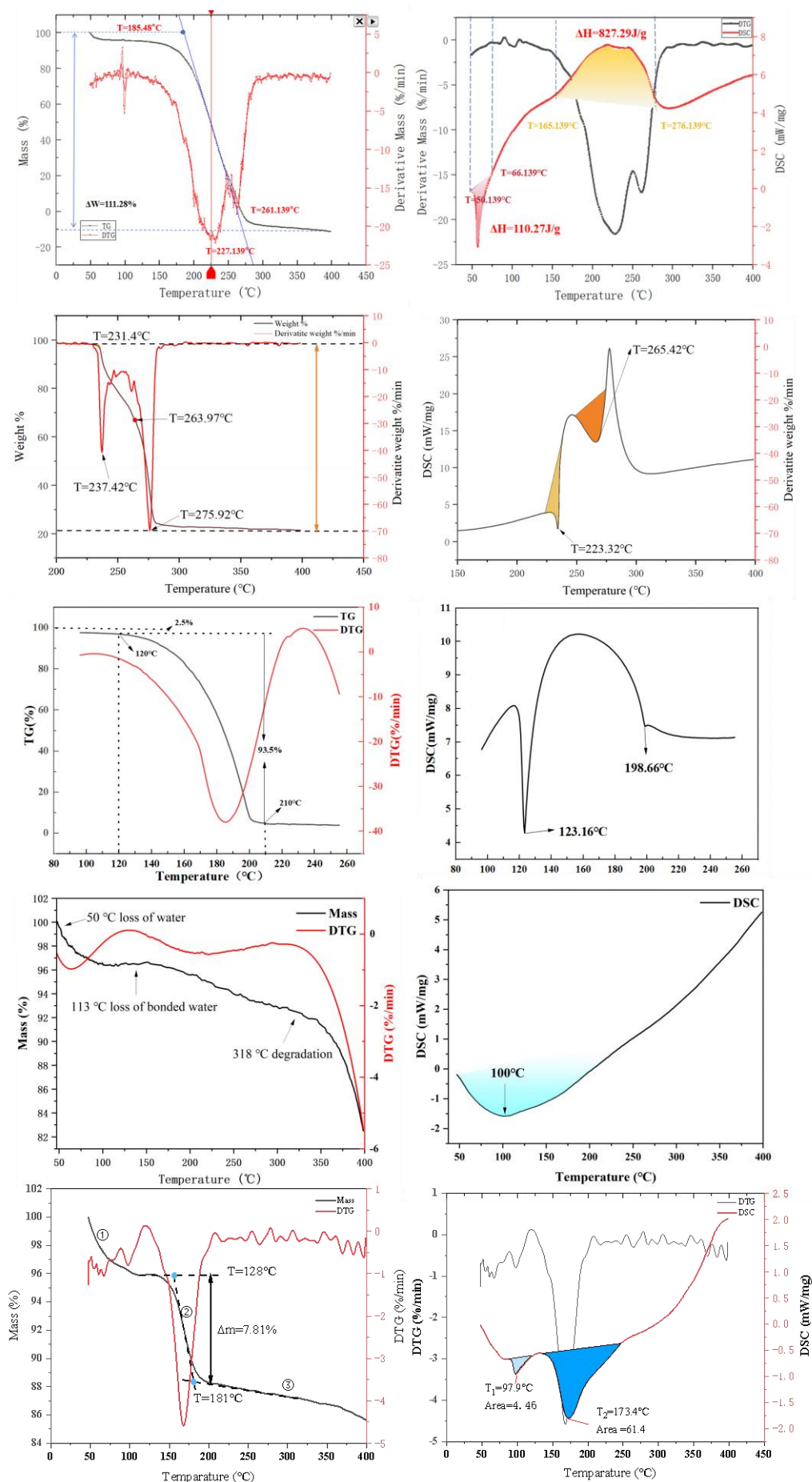


Figure 2. The Thermogravimetric analysis spectrum of the samples

2. Infrared Spectroscopy

2.1 Infrared Ray

Infrared is an electromagnetic wave with a wavelength range of $0.76\mu\text{m}$ - $1000\mu\text{m}$. When irradiated by this light source, it could produce radiation, penetration, absorption and resonance on the organism^[1]. Infrared light generates heat by resonating with inorganic molecules and organic macromolecules, accelerating the movement of these molecules and rubbing them against each other. Infrared radiation can be categorized into three sections: Short-infrared (high-frequency infrared with higher energy), with wavelengths from $1.0\mu\text{m}$ to $0.75\mu\text{m}$; Mid-infrared (mid-frequency infrared with moderate energy), with wavelengths from $40\mu\text{m}$ to $2.5\mu\text{m}$; and Far-infrared (low-frequency infrared with lower energy), with wavelengths from $1500\mu\text{m}$ to $25\mu\text{m}$. Moreover, owing to dual nature of infrared and microwave, IR has received much attention in scientific research and widely applied in biology, chemistry, molecular spectroscopy, organic synthesis and other subject areas.

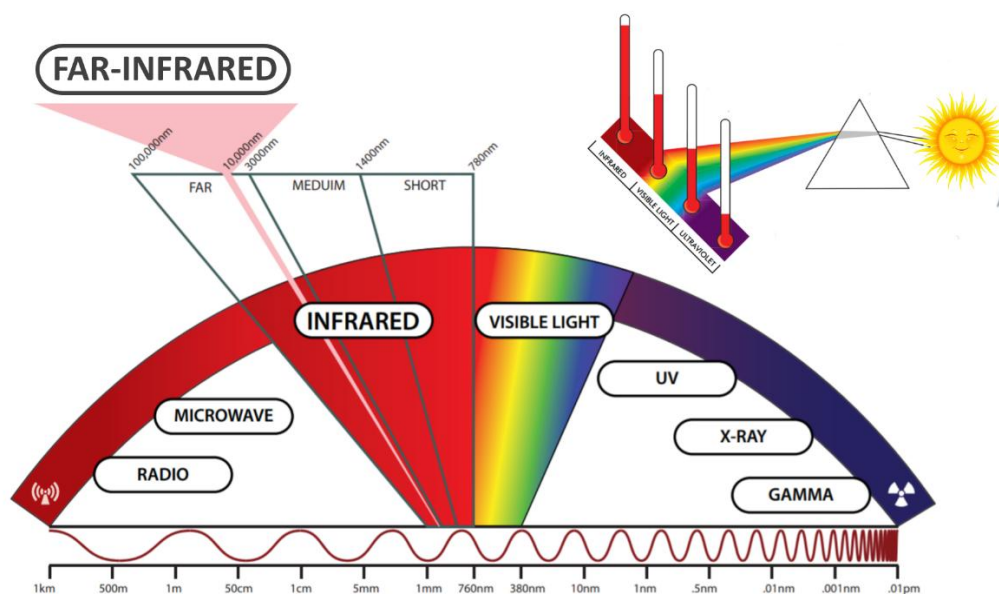


Figure 3. Position of infrared ray in the spectrum^[2].

2.2 Infrared Spectrometer

Infrared spectroscopy is a tool for determining the molecular structure and chemical composition of substances based on information about atom vibration and molecular rotation. Infrared spectroscopy has advantages of rapid testing, ease of operation, minimal sample requirement, and high sensitivity, therefore, it is indispensable in studies of polymer configuration, conformation, mechanical properties, as well as in the fields of physics, biology, and medicine.

2.2.1 Principle

When a beam of infrared light with continuous wavelengths passes through a substance, functional groups with the same vibrational (rotational) frequency undergo transitions from the ground vibrational (rotational) energy level to higher energy levels. The light at these wavelengths is absorbed by the substance. Different chemical bonds or functional groups have different absorption frequencies and thus appear at different positions in the infrared spectrum.

IR spectroscopy working principle is shown in Fig 4. Initially, the light source emits infrared light across a broad spectrum of wavelengths. The infrared light then passes through the monochromatic system, which filters it into a beam of monochromatic light. Subsequently, the monochromatic light is split by a splitter into two separate beams, one acting as a reference beam and the other as the sample beam. As these beams interact with the sample, the sample absorbs infrared light at specific wavelengths. The resulting difference in intensity between the two beams is measured by the detector, with this variation being due to the absorption of infrared light at specific wavelengths by the sample.

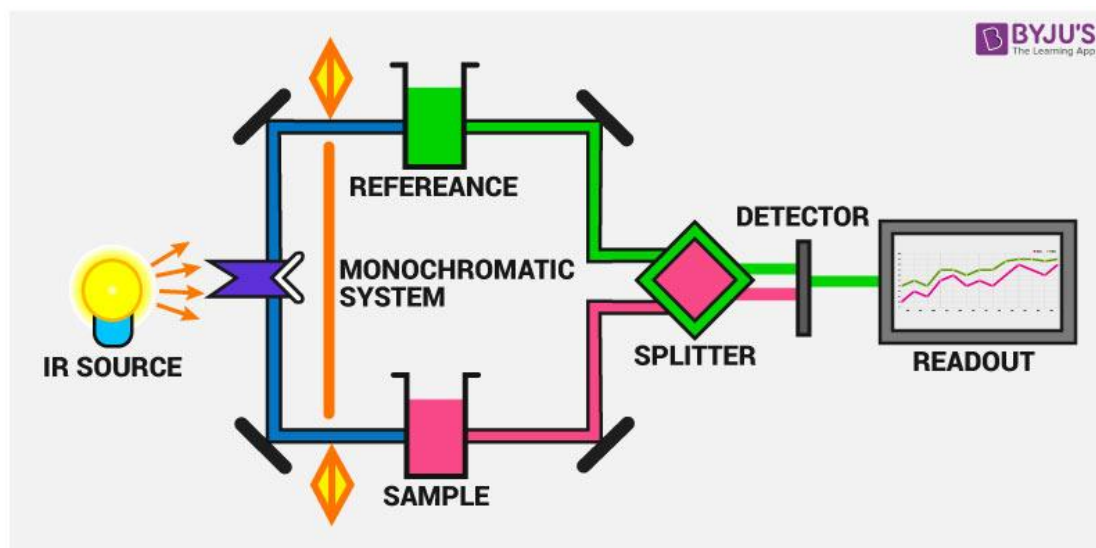


Figure 4. IR spectroscopy working principle^[3].

The generation of an infrared absorption spectrum requires two major conditions: (a) the energy of the electromagnetic wave must be equal to the difference between the two molecular energy levels. This determines the position where the absorption peaks occur. (b) there must be an odd interaction between the infrared light and the molecules. Therefore, when molecules vibrate, their dipole moment must change.

2.2.2 Absorbance

Infrared Spectroscopy Absorbance refers to the ability of a substance to absorb infrared light with specific wavelengths during infrared spectral analysis. It is commonly used to quantify the extent of a sample's absorption of infrared light and is an important parameter in infrared spectroscopy. Infrared absorption spectroscopy can provide information about the chemical structure and composition of a substance.

2.2.3 Vibration modes

The essential difference between infrared and Raman spectroscopy is that the activity of the infrared spectrum depends on whether the dipole moment changes. IR vibrations can be divided into two categories [4]: bending and stretching, stretching can be divided into symmetric stretching and asymmetric stretching; bending can be divided into out-of-plane bending and in-plane bending, out-of-plane bending with wagging and twisting, and in-plane bending with rocking and twisting.

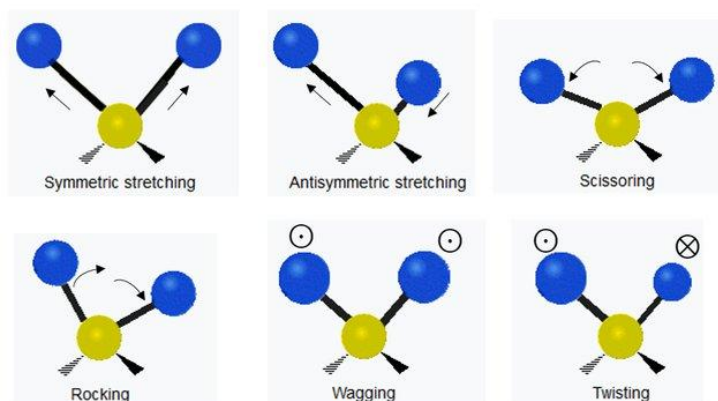


Figure 5. IR spectroscopy working principle [5].

2.2.4 Equipment used in the experiment

An infrared spectrometer (as shown in Figure 6) is an instrument that analyzes the molecular structure and chemical composition of substances by utilizing their absorption characteristics of infrared radiation at different wavelengths. An infrared spectrometer typically consists of a light source, a monochromator, a detector, and a computer information processing system. Based on the type of dispersing device, it can be categorized into dispersive and interferometric types(1)[6].



Figure 6 . Invenio German Brooke infrared spectrometer(1)[6]

For dispersive dual optical path optical zero balance infrared spectrophotometer, when the sample absorbs infrared radiation of a certain frequency, the molecular vibration energy level transitions, and the light of the corresponding frequency in the transmitted beam is weakened, resulting in the intensity difference between

the corresponding radiation of the reference optical path and the sample optical path, thus obtaining the infrared spectrum of the measured sample(1).

2.3 Experimental Procedure

1. Sample preparation: Grind the solid sample to make it become fine powder.
2. Instrument warm up: Turn on the IR spectrometer and allow it to warm up for some time to ensure that the light source and detector reach a stable state.
3. Instrument calibration: Instrument is calibrated using known reference materials to ensure accuracy of wavelength and intensity.
4. Background scanning: Scan is performed in the absence of a sample and the background spectrum is recorded.
5. Sample testing: Place the sample evenly on the sample stage.
6. Data acquisition: Start the scanning program and let the IR spectrometer collect spectral data from the sample.
7. Data processing: Collected spectral data is compared to the background spectrum to eliminate the effect of the background.

2.4 Attenuated Total Reflectance (ATR)

Compared to the transmission KBr press test, the Attenuated Total Reflectance (ATR) technique requires almost no treatment of the sample during the testing process and causes no damage to the sample, making it an important means of Fourier Transform Infrared Spectroscopy (FTIR) testing. It is a convenient way to test solid, liquid and thin film samples in a very short period, without the need for sample preparation. The working principle of the ATR accessory is described below.

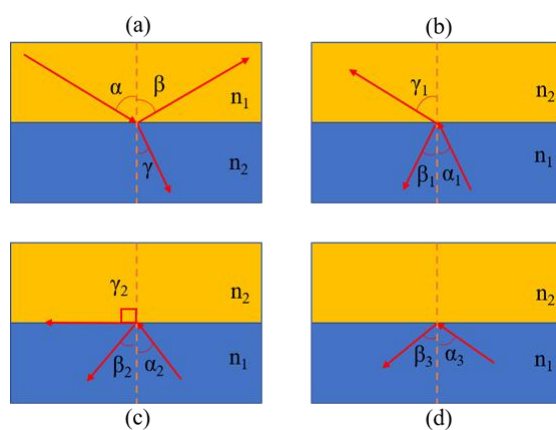


Figure 7. NETZSCH Fourier Transform Infrared (FTIR) spectroscopy(1)

The basic principle of ATR technology is to bring the sample into contact with a crystal disk that has a high refractive index.(2) When infrared light passes through the crystal disk and enters the sample, due to refraction and reflection, the light forms total internal reflection at the sample surface. In the state of total internal reflection, the light interacts with the sample, a portion of the light intensity can penetrate the sample, and is detected by a detector, thereby obtaining the sample's infrared spectral information.

2.5 Fourier transform infrared spectroscopy (FTIR)

Fourier Transform Infrared Spectroscopy (FTIR) is a technique for analyzing infrared spectra using the Fourier Transform technique, which measures the molecular absorption spectra formed by molecular groups absorbing radiation of specific frequencies by irradiating continuous wavelengths of infrared light onto the sample.



Figure 8. NETZSCH Fourier Transform Infrared (FTIR) spectroscopy

Fourier transform infrared spectroscopy is characterized by high signal-to-noise ratio, high resolution and high sensitivity, fast scanning speed and wide spectral range for solid, liquid or gas samples. It is suitable for analyzing solid, liquid or gas samples. It has a wide range of applications in the fields of material science, biology, medicine and environment, including group structure analysis, qualitative and quantitative analysis, cure rate testing, environmental monitoring and industrial process monitoring, and plays an important role in modern analytical chemistry because of its high efficiency and wide applicability.

3. Infrared spectroscopic analysis

3.1 Sample A1-analyzed by Zherui Hou

3.1.1 Structural analysis

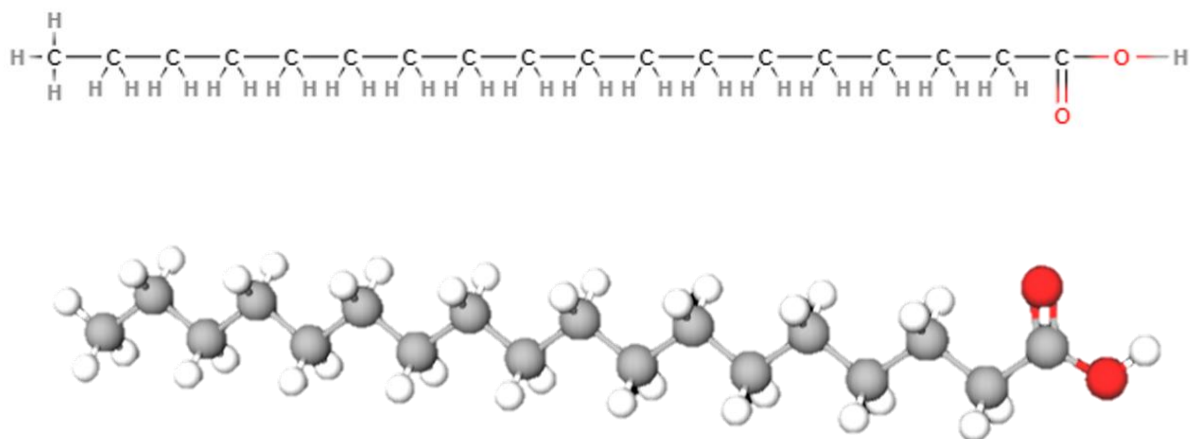


Figure 9. Molecular structure and molecular modeling of stearic acid

The chemical structure of stearic acid includes C=O, C-O, and long-chain alkanes.

3.1.2 Infrared analysis

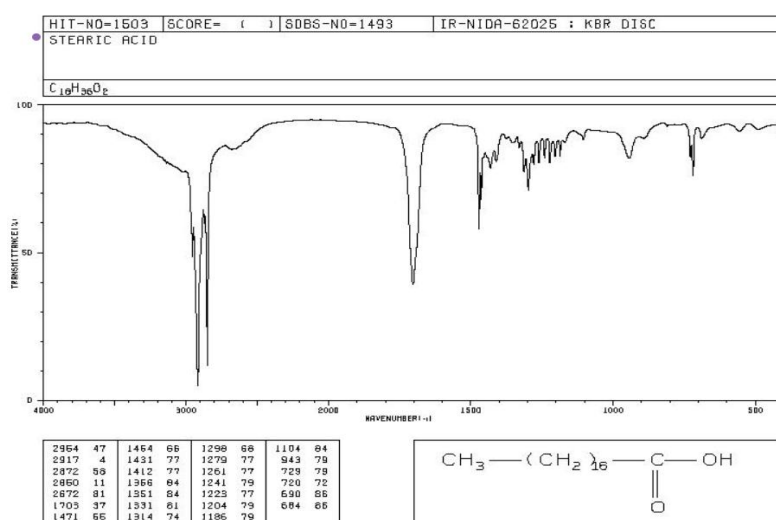


Figure 10. Standard infrared spectrum of stearic acid

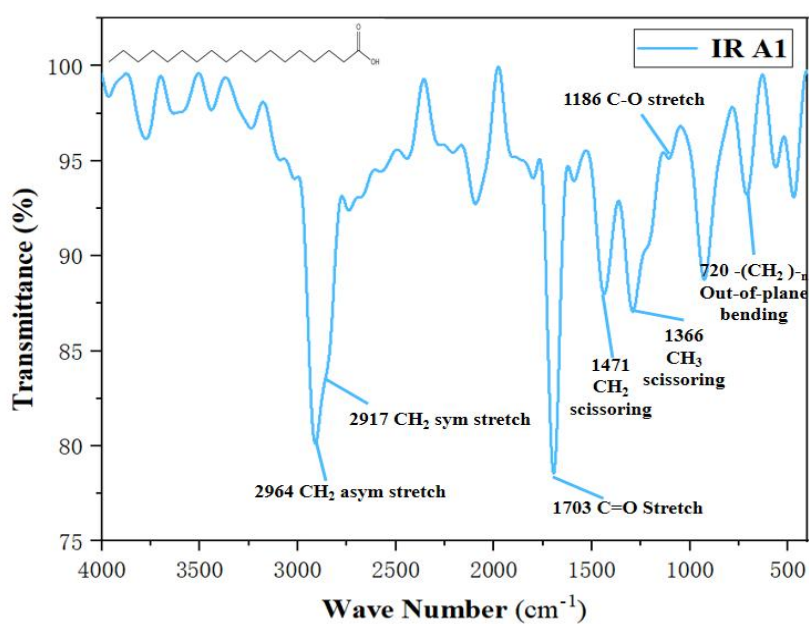


Figure 11. Infrared data plot of sample A1

By comparing the infrared spectrum of the drawn sample with the standard spectrum of stearic acid, it can be found that the two not only have similar shapes, but also fit well in the wavenumber of characteristic peaks. The asymmetric and symmetric stretching vibrations of methylene (CH_2) at 2964cm^{-1} and 2917cm^{-1} , the stretching vibration of carbonyl at 1703cm^{-1} , and the bending vibrations of methylene and methyl at 1471cm^{-1} and 1366cm^{-1} , respectively, are speculated to be the scissoring vibration mode. The stretching vibration of C-O bond at 1186cm^{-1} and the bending vibration of n methylene homopolymers at 720cm^{-1} can all correspond to the characteristic peaks on the standard spectrum. Therefore, it is determined that sample A1 is stearic acid.

But we also found that there are some small peaks around 3500cm^{-1} that do not belong to the characteristic peaks of stearic acid. We infer that these peaks are due to factors such as moisture during the sample preparation process, with water involved, resulting in symmetric and asymmetric stretching vibrations of O-H bonds.

It is worth discussing here that the stretching vibration of C=O bonds in general carboxylic acids is generally around 1710cm^{-1} , and here the hydroxyl group, which is a strong electron withdrawing group, is connected. According to the induction effect, the stretching vibration frequency of C=O bonds should increase, but this effect is also reduced due to the large number of long-chain alkyl groups, which are electron repelling groups, connected to the other side.

Table 2. Infrared spectral analysis

Peak	Wavenumber(cm^{-1})	Bond/Functional Goup	Vibrational mode
1	2964	CH_2	Asym stretching
2	2917	CH_2	Sym stretching
3	1703	C=O	Stretching
4	1471	CH_2	Scissoring
5	1366	CH_3	Scissoring
6	1186	C-O	Stretching
7	720	$-(\text{CH}_2)_n$	Out-of-plane bending

3.2 Sample A2 - analyzed by Chengrui Yang (2022304152)

3.2.1 Structural analysis

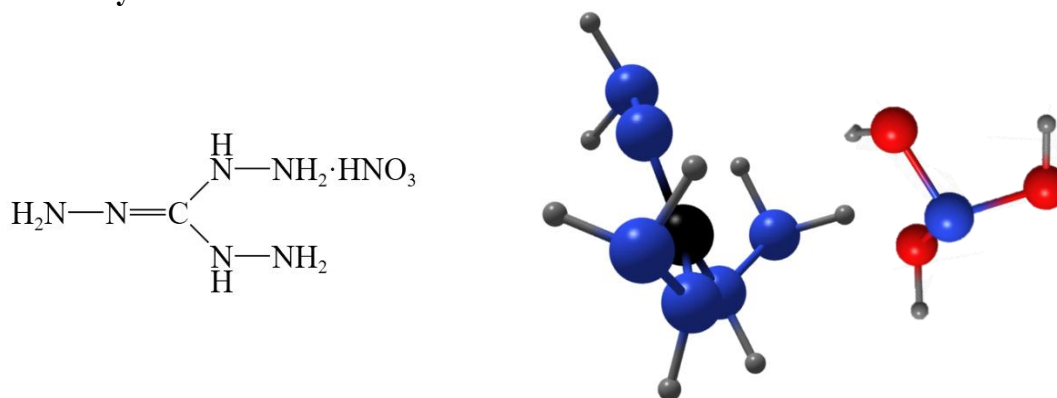


Figure 12. Molecular structure and molecular modeling of TAGN.

The chemical structure formula of urea shows that its structure has C-N bond, N-H bond, N=C bond, N-O bond.

3.2.2 Standard infrared analysis

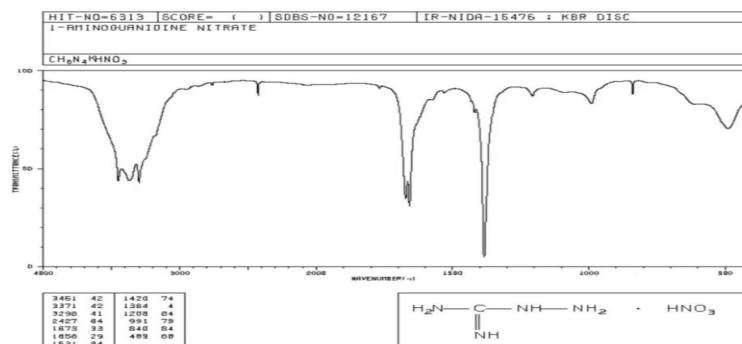


Figure 13. Standard infrared absorption spectra of TAGN[6].

Through the analysis of standard infrared spectra, it can be predicted that there should be N-H, N-O symmetric and asymmetric stretching vibration, C=N, C-N stretching vibration and N-H, N-O, bending vibration in the infrared detection spectrum of area.

3.2.3 Infrared analysis

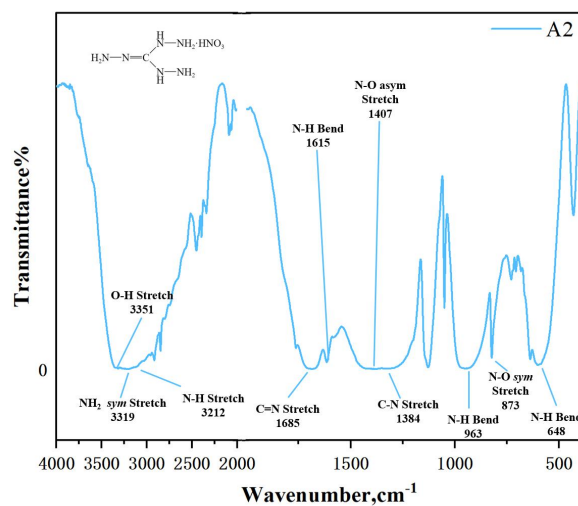


Figure 14. Infrared absorption spectra of TAGN.

In the infrared spectrogram of TAGN, from group frequency region, it shows that the transmittance peaks at 3319 cm^{-1} and 3212 cm^{-1} are the symmetric stretching and stretching vibration of N-H(amide I) and N-H(amide II). The transmittance peaks at 3351 cm^{-1} could be O-H stretching. The transmittance peak at 1685 cm^{-1} is the stretching vibration of C=N and the transmittance peak at 1615 cm^{-1} are the bending vibration of N-H (amide II). The transmittance peaks at 1407 cm^{-1} are the stretching vibration of N-O. C-N stretching peak can be seen in 1384 cm^{-1} .

Moreover, in the fingerprint region part, the transmittance peak at 963 cm^{-1} is N-H (amide II) bending and transmittance peak at 648 cm^{-1} is N-H(amide I) bending. And the peak at 873 cm^{-1} is the N-O symmetric stretching.

The unlabeled peaks were thought to water or impurities disruptions during the experimental manipulation. The results are shown in the following table:

Table 3. Infrared spectral analysis

Peak	Wavenumber(cm^{-1})	Bond type	Vibration mode
1	3351	O-H	Stretching
2	3319	N-H	Symmetric Stretching
3	3212	N-H	Stretching
4	1685	C=N	Stretching
5	1615	N-H	Bending
6	1407	N-O	Asymmetric Stretching
7	1384	C-N	Stretching
8	963	N-H	Bending
9	873	N-O	Symmetric Stretching
10	648	N-H	Bending

3.3 Sample A3--analyzed by Bokun Zhao(2022304147)

3.3.1 Infrared analysis of benzoic acid

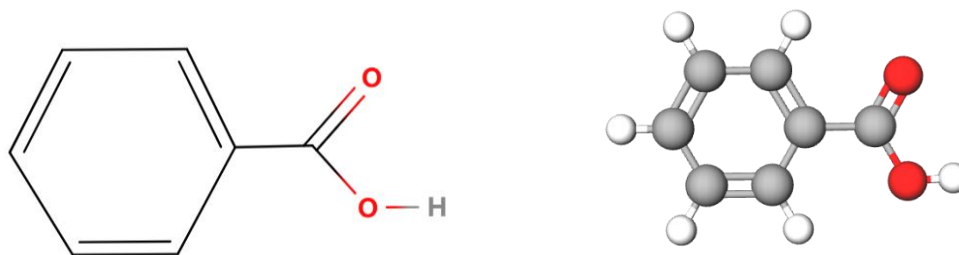


Figure 15 . Chemical structure formula of benzoic acid

By looking at the chemical structure formula of benzoic acid, the chemical bonds that can be obtained are: C-O single bond, C=O double bond, O-H bond, C-H bond, benzene, and vibration modes such as bending and stretching vibrations.

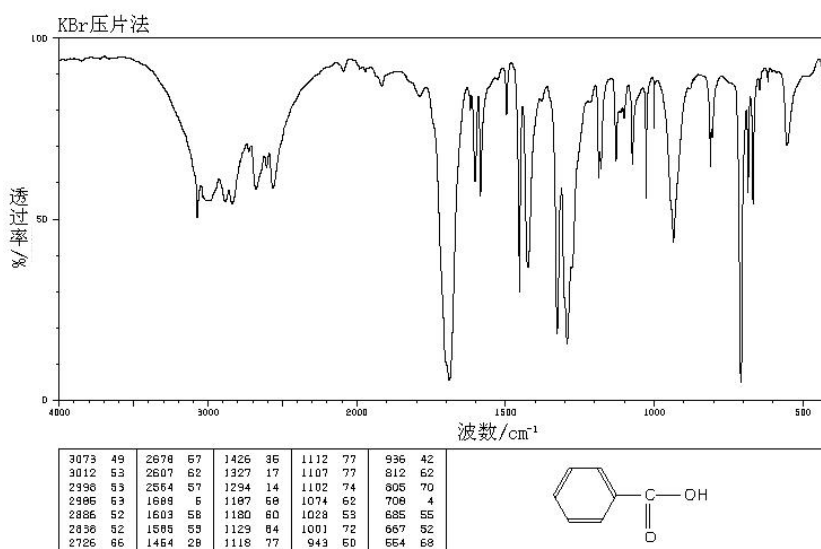


Figure 16. Standard infrared absorption spectra of benzoic acid[1]

As we can see in Figure 17, 3662 cm⁻¹ is absorption peak of O-H stretching, 1687 cm⁻¹ is absorption peak of C=O stretching, 1603 cm⁻¹、1582cm⁻¹、1496cm⁻¹、1454 cm⁻¹ are absorption peaks of benzene ring vibration, 1289 cm⁻¹ and 1182 cm⁻¹ are absorption peaks of C-O stretching, 1072cm⁻¹ is absorption peak of C-H in plane bending, 932-710cm⁻¹ is absorption peak of C-H out plane bending.

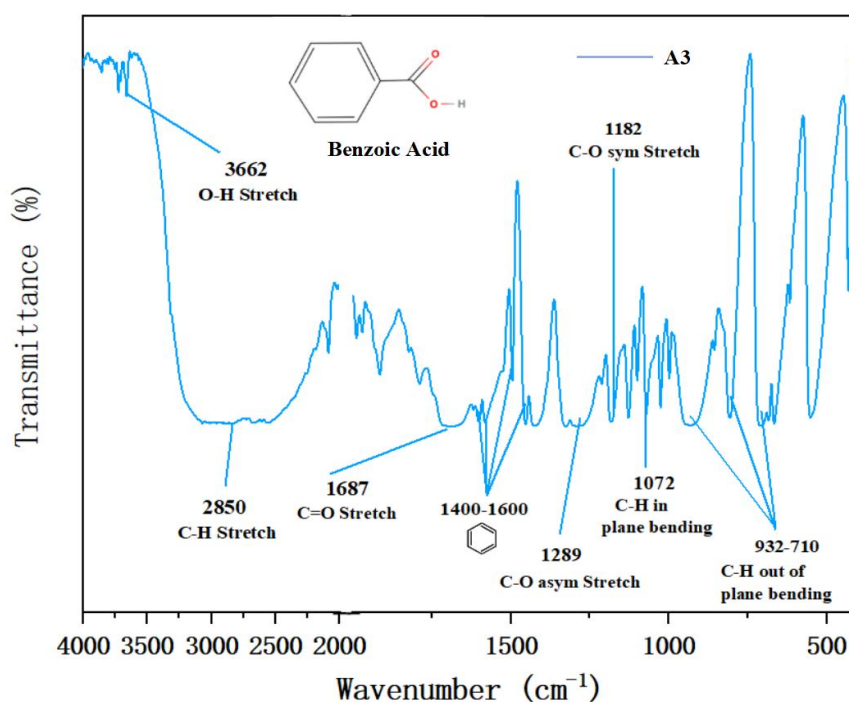


Figure 17. Infrared spectrum of benzoic acid

Table 4. Infrared spectral analysis of benzoic acid

Peak	Wavenumber(cm^{-1})	Bond type	Vibration mode
1	710	C-H	Out of plane bending
2	856	C-H	Out of plane bending
3	932	C-H	Out of plane bending
4	1072	C-N	In plane bending
5	1289	C-O	Asymmetric Stretching
6	1454	Benzene ring	Benzene ring vibration
7	1496	Benzene ring	Benzene ring vibration
8	1582	Benzene ring	Benzene ring vibration
9	1603	Benzene ring	Benzene ring vibration
10	1687	C=O	Stretching
11	2850	C-H	Stretching

3.4 Sample A 4 - analyzed by Zhuoyu Wang (2022304138)

3.4.1 Structural analysis

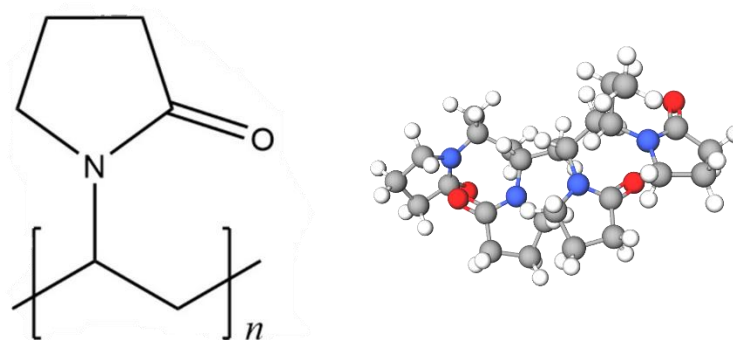


Figure 18. Molecular structure and molecular modeling of Polyvinyl pyrrolidone.

The chemical structure formula of Polyvinyl pyrrolidone shows that its structure has C-N single bond, C-C single bond, C-H bond and C=O double bond.

3.4.2 Standard infrared analysis

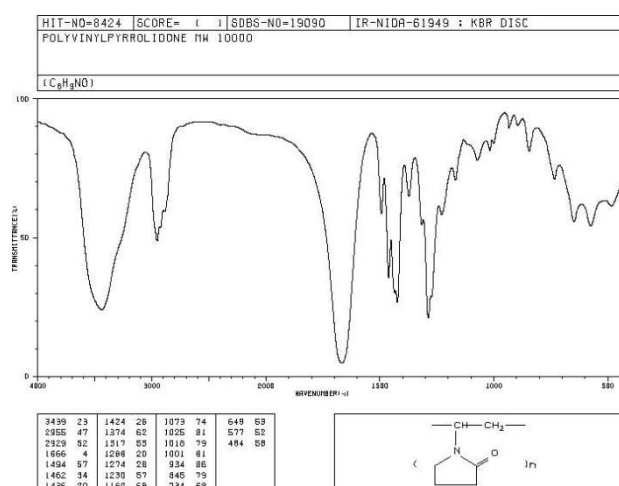


Figure 19. Standard IR spectrum of Polyvinyl pyrrolidone

Through the analysis of standard infrared spectra, it can be predicted that there should be C-H asymmetric stretching and symmetric stretching, C-H stretching affected by N atom and C=O group, C=O stretching affected by the carbon ring and N atom, C-C stretching, C-N stretching, C-H asymmetric stretching, symmetric stretching and bending.

3.4.3 Infrared analysis

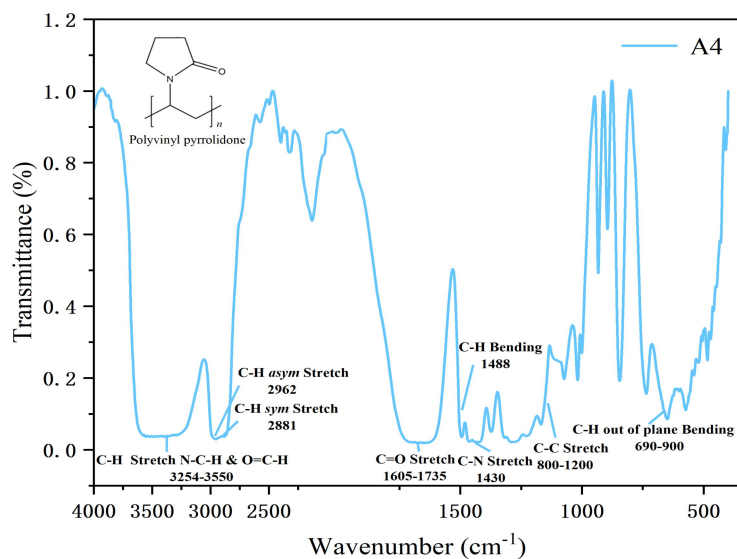


Figure 20. IR spectrum of Polyvinyl pyrrolidone

The results are shown in the following table:

Table 5. Infrared spectral analysis

Peak	Wavenumber(cm ⁻¹)	Bond type	Vibration mode
1	3254-3550	C-H (attach to N or C=O group)	Stretching
2	2962	C-H	Asymmetric stretching
3	2881	C-H	Symmetric stretching
4	1605-1735	C=O	Stretching
5	1488	C-H	Bending
6	800-1200	C-C	Stretching
7	C-H	690-900	Out of plane bending

3.5 Sample A5 - analyzed by Zeying Jin(2022304167)

3.5.1 Structural analysis

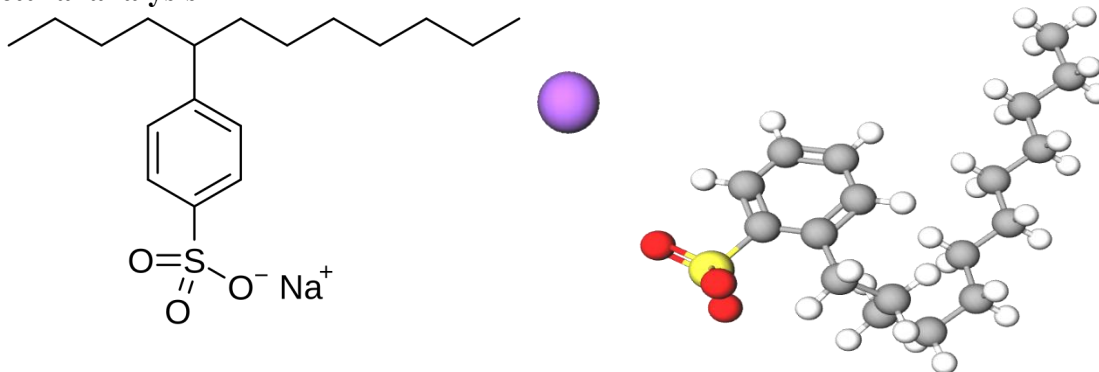


Fig 21. Molecular structure and molecular modeling of SDBS.

The chemical structure formula shows that its structure has Benzene ring, $-(CH_2)_n$, Sulfonate ($O^- - S(=O)_2 - R'$) and S-C bonds

3.5.2 Standard Infrared Analysis

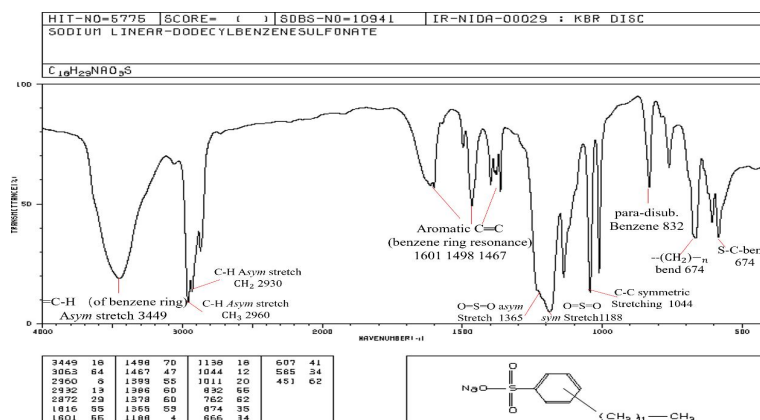


Figure 22. Standard IR absorption spectra of SDBS [13].

Through the analysis of standard infrared spectra, it can be predicted that there should be =C-H (of benzene ring) Asymmetric stretching, C-H Asymmetric stretching for -CH₂ and -CH₃, Aromatic C=C from Benzene ring resonance, R-S(=O)₂-R' Asymmetric stretching and symmetric stretching, C-C symmetric stretching, para-disub. Benzene, saturated hydrocarbon with long carbon chains, -(CH₂)_n (n>4), presents 722 cm and decreases with the increase of n, at n = 11, $\tilde{\nu}$ decreases to 674. And S-C bending^[1].

3.5.3 Infrared analysis

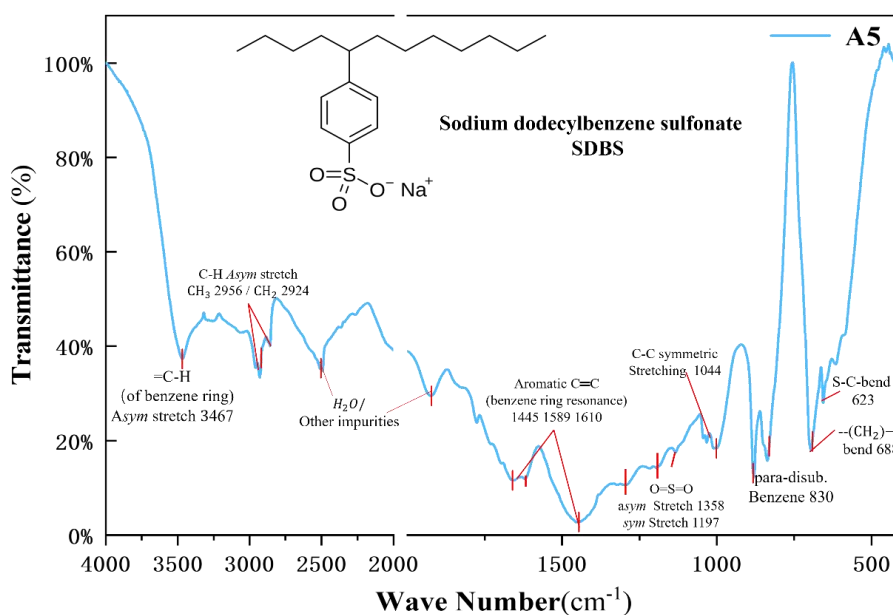


Figure 23. Infrared absorption spectra of SDBS.

In the infrared spectrum of sample A5 tested in this experiment, first of all, in the functional group region, there is peak 1 with a frequency of 3467. At first, we consider the asymmetric stretching of =C-H in the benzene ring. Normally, its frequency range should be around 3030. However, the sulfonate is a group with quite strong electronegativity. Due to the electron-withdrawing effect (-ve inductive effect) in the inductive

effect, the benzene ring serves as an electron-donating group. As a result, the bond length of the C-H bond gets shorter, the force constant K becomes larger, and eventually the frequency increases significantly. Besides, the distances of C-H at different positions on the benzene ring vary, so there is a certain range around 3467. The saturated alkane chains of peak 2 and 3 belong to CH_2 and CH_3 respectively, with frequencies of 2956 and 2924, both of which are asymmetric stretching. For peak 4, compared with the standard infrared spectrum in Figure 5.1, there is no such functional group, so it is regarded as water or other impurities. For peak 5, we can see frequencies of 1445, 1589 and 1610, which correspond to the Aromatic C=C bending in the benzene ring skeleton vibration. As for peak 6 and 7, they correspond to the asymmetric and symmetric stretching of sulfonate respectively. For peak 8 with a value of 1044, it can be the symmetric stretching of the C-C bond. The value of peak 9 is 830, which exactly matches the para-disubstituted C-H bending of the benzene ring. For peak 10 with a value of 688, since the saturated alkane chain has the maximum frequency of 722 when $n = 4$, and in this experiment the sample has $n = 11$, so the frequency is reduced to 688. Finally, the 623 of peak 11 can be considered as the bending vibration of the S-C bond.[2]

The results are shown in the following table:

Table 6. IR Peak Marking

Peak	Wavenumber(cm^{-1})	Bond type	Vibration mode
1	3467	=C-H	Asymmetric Stretching
2	2956	C-H (CH_3)	Asymmetric Stretching
3	2924	C-H (CH_2)	Asymmetric Stretching
4	2505/1921	Unknown	Unknown
5	1445 1589 1610	Aromatic C=C	Bending benzene ring resonance
6	1358	$\text{O}^- - \text{S}(=\text{O})_2 - \text{R}'$	Asymmetric Stretching
7	1197	$\text{O}^- - \text{S}(=\text{O})_2 - \text{R}'$	Symmetric Stretching
8	1044	C-C	Symmetric Stretching
9	830	C-H	Bending para-disub. Benzene (1,4-disubstituted)
10	688	C-H ($-(\text{CH}_2)_n$)	Bending
11	623	S-C	Bending

3.6 Sample A 6.0 - analyzed by Yuxin Liu (2022304143)

3.6.1 Structural analysis

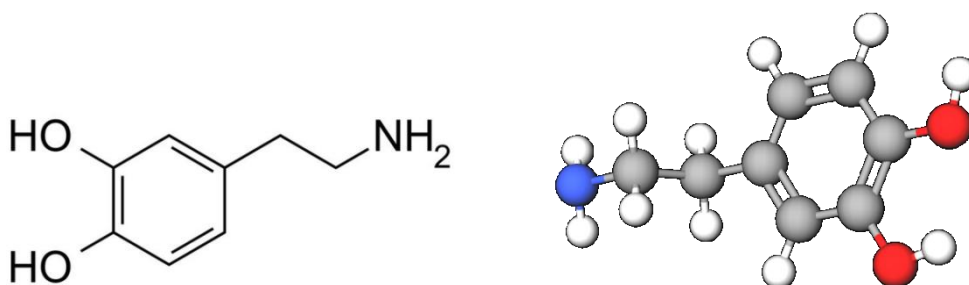


Figure 24. The schematic of the Dopamine (a). Structural formula; (b). Ball-and-stick model

The chemical structure formula of Dopamine shows that its structure has hydroxyl, amino function groups and benzene ring. And there probably exist hydrogen effect caused by above two function groups and conjugation effect caused by the benzene structure.

3.6.2 Infrared analysis

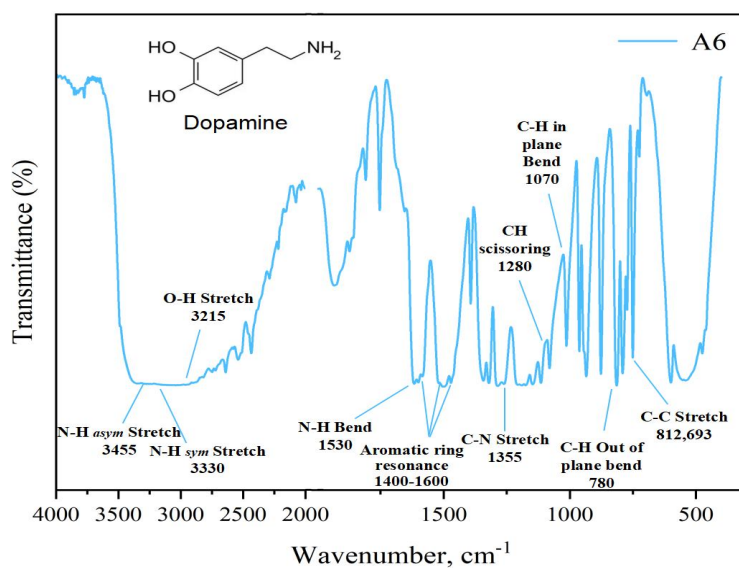


Figure 25. The Infrared spectrum of Dopamine in the experiment

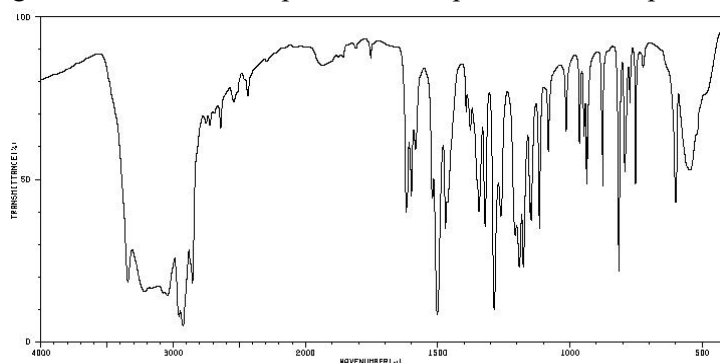


Figure 26. The Standard Infrared spectrum of Dopamine^[1]

Through the analysis of the infrared spectrum of Dopamine in the experiment and compare the data with the standard infrared spectrum. It could be find that there exist a very big peak in the range of 2800-3500 cm⁻¹, which might be the mixed peak of aysm Stretch N-H bond, sym Stretch N-H bond and Stretch O-H bond. It may also be due to the sample delixing during the detection process, resulting in the interference of water molecules in the detection process, making the peak width larger and more superimposed of O-H stretching at the same time^[2]. And it's worth noting that the stretch O-H bond has a relative low frequency because of the conjugation effect by connecting with the benzene ring^[3]. There also exist typical aromatic ring resonance at 1400 to 1600 cm⁻¹ and C-N stretch at 1355 cm⁻¹. Due to the presence of multiple substitution sites on the benzene ring, the spectrum of the fingerprint region is complicated and consist of the C-H scissoring at 1280 cm⁻¹, C-H out of plane bend at 780 cm⁻¹ and C-C stretch at 812 and 693 cm⁻¹.

Generally, the standard spectrum is consistent with the experimental spectral data to some extent. But there are also problems such as different peak width and peak intensity, which maybe caused by the presence of impurities in the sample in the experiment, water absorption of the sample in the air, and the intrinsic problem of the detection equipment.

Table 7. Infrared spectral analysis of Dopamine

Peak	Wavenumber(cm ⁻¹)	Bond type	Vibration mode
1	3455	N-H	<i>aysm</i> Stretch
2	3330	N-H	<i>sym</i> Stretch
3	3215	O-H	Stretch
4	1555	N-H	Bend
5	1355	C-N	Stretch
6	1280	C-H	Scissoring
7	780	C-H	Out of plane bend
8	812,693	C-C	Stretch

4. Introduction of Thermal Analysis

Thermal analysis is a common method in scientific research characterization. The so-called thermal analysis refers to the change of the corresponding physical and chemical properties of the sample by controlling the change of temperature. The three most common methods of thermal analysis are

thermogravimetry (TG), differential thermal analysis (DTA), and differential scanning calorimetry (DSC). In this report, TG and DSC will be introduced in detail:



Figure 27. The schematic of TGA and DSC

4.1 TGA

4.1.1 Introduction

The mass (m) of the sample is measured with temperature under programmed temperature control. The TG curve can be used to understand the changes in sample quality (such as adsorption, desorption, decomposition, etc.) during the heating or cooling process of the sample. For example, there is often carbon deposition in industrial catalysts, and the carbon deposition amount can be determined through TG characterization. In the analysis of TG images, the most important information is the temperature point of weightlessness and the proportion of weightlessness. Based on the properties of the material tested and these temperature points and weight loss ratios, the physicochemical changes that occur can be inferred^[8].

There are two methods commonly used in TGA thermogravimetric analyzer: static method and dynamic method. The static method includes isobaric mass change measurement and isothermal mass change measurement. Isobaric mass change measurement is a method to measure the relationship between mass and temperature of a substance at a constant partial pressure of a volatile substance at a programmed temperature. Isothermal mass change measurement is a method to measure the relationship between mass and pressure under constant temperature conditions^[9]. The static method is accurate but time-consuming. The dynamic rule is what we often call thermogravimetric analysis and derivative thermogravimetric analysis.

Thermogravimetric analysis is the first derivative of TG curve to temperature (or time), and the DTG curve is obtained by plotting temperature T (or time t) with the mass change rate of a substance.

4.1.2 Instruments

In thermo-gravimetric analysis (TGA) instruments, the temperature control system, the detection system, and the recording system are the key components, which are used to achieve temperature control, the detection of sample quality changes and the recording and analysis of data, respectively.



Figure 28. Thermogravimetric analysis experiment using instruments. (NETZSCH)

4.1.3 Experiment procedures

4.1.3.1 Setting test parameter:

Heating rate: 15°C/min

Temperature range: from 50°C to 400°C

Atmosphere condition: Argon atmosphere(about 40 mL/min)

Crucible type: Alumina crucible

4.1.3.2 Sample preparation:

1) The mass of the sample must less than 10mg and can be presented as solid, liquid, powder, film or fiber samples. The bulk sample should be grinded to the size less than 4mm in diameter and less than 0.5mm in thickness.

2) The composition of the sample must be clearly labeled.

3) Place the sample to be tested in the crucible with the reference sample after weight measurement.

4) Set the program and start heating.

5) The thermo-gravimetric analysis of the sample is completed, the instrument temperature drops below 50 degrees Celsius before opening the instrument and taking out the sample.

4.1.3.3 Data processing

After the test is completed, the experimental data and export the TGA, DTG and DSC curve. Analyze the data as required including the enthalpy/heat flow, phase change, glass transition, heat capacity, mass changes and coefficient of heat expansion of the sample. Finally plot the TG, DTG and DSC curve and compare them with the literature information to determine the type of the sample.

4.2 DSC

4.2.1 Definition

Differential Scanning Calorimetry (DSC) is a thermodynamical tool for direct assessment of the heat energy uptake, which occurs in a sample within a regulated increase or decrease in temperature. DSC

measures the heat flux to or from a sample specimen as well as enthalpy changes as a function of temperature or time.

4.2.2 Classification

4.2.2.1 Heat-Flux DSC

Heat-Flux DSC is a thermal analysis technique that quantifies the difference in heat flow between a sample and a reference material as they are subjected to the same temperature program. This method allows for the precise determination of a sample's heat capacity and phase transition enthalpies.

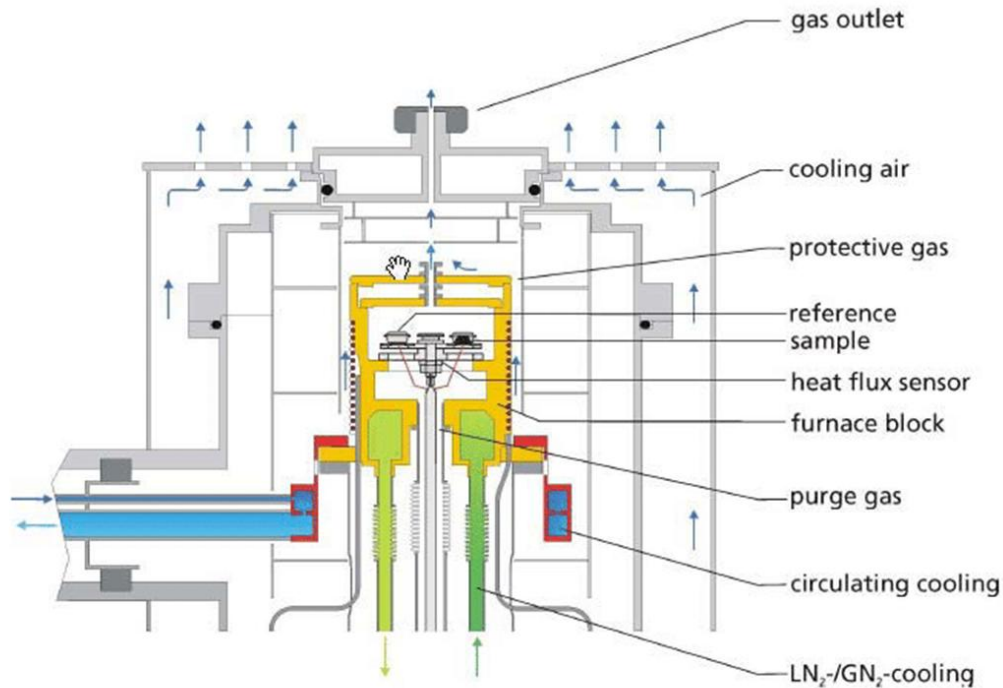


Figure 29. Schematic diagram of heat flow DSC

Principle and Formula:

The underlying principle of heat-flux DSC is to quantify the thermal flow disparity between a substance and a reference as they are subjected to the same temperature program. The heat capacity of the sample, C_{ps} , can be derived from the following formula:

$$C_{ps} = \frac{C_{pr} \cdot (m_r \cdot \Delta T_m - m_r \cdot \Delta T_r - m_s \cdot \Delta T_s)}{m_s \cdot (\Delta T_m - \Delta T_r)}$$

where ΔT_m , ΔT_r , and ΔT_s represent the temperature differences for the empty pans, the reference sample, and the sample itself. m_r and m_s denote the masses of the reference and the sample, respectively, and C_{pr} is the heat capacity of the reference material^[14].

4.2.2.2 Power Compensation DSC

Power Compensation DSC maintains the sample and reference at the same temperature and measures the power differential required to sustain this condition. This method is particularly adept at capturing detailed thermal effects of the sample, such as phase transitions and reaction enthalpies.

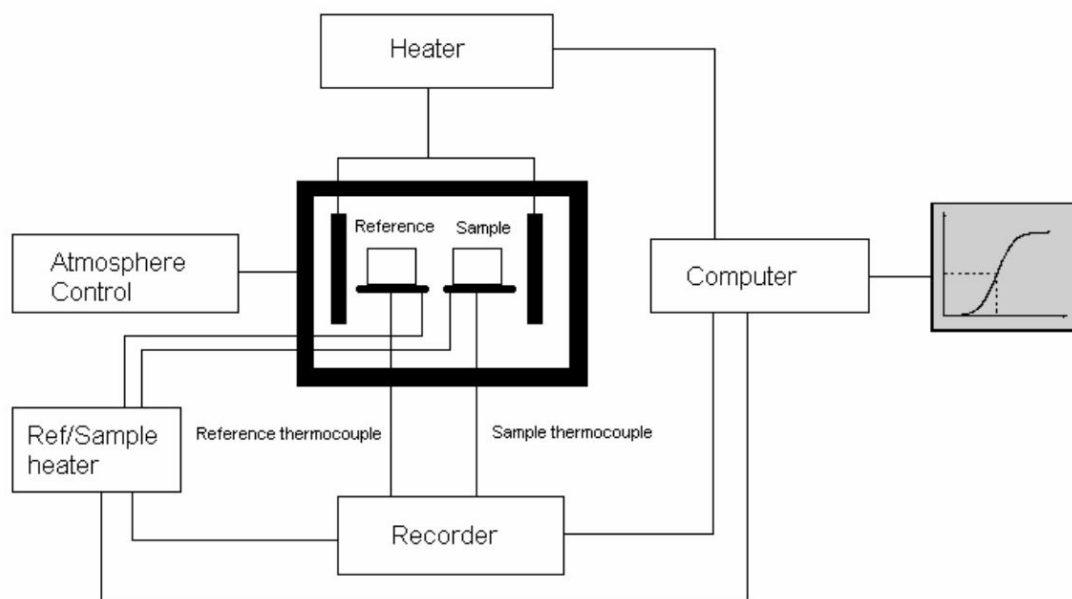


Figure 30. Schematic diagram of compensating DSC

Principle and Formula:

The power compensation DSC is designed to measure directly and exactly the enthalpy change of the sample holder as a function of time. The power difference, ΔP , between the sample and reference can be expressed by the equation:

$$\Delta P = \frac{\Delta H}{\Delta T}$$

Where ΔH and ΔT represent the change in enthalpy and temperature, respectively^[15].

5. Thermal Analysis

5.1 Sample A1

5.1.1 Analyze of TG and DTG curve

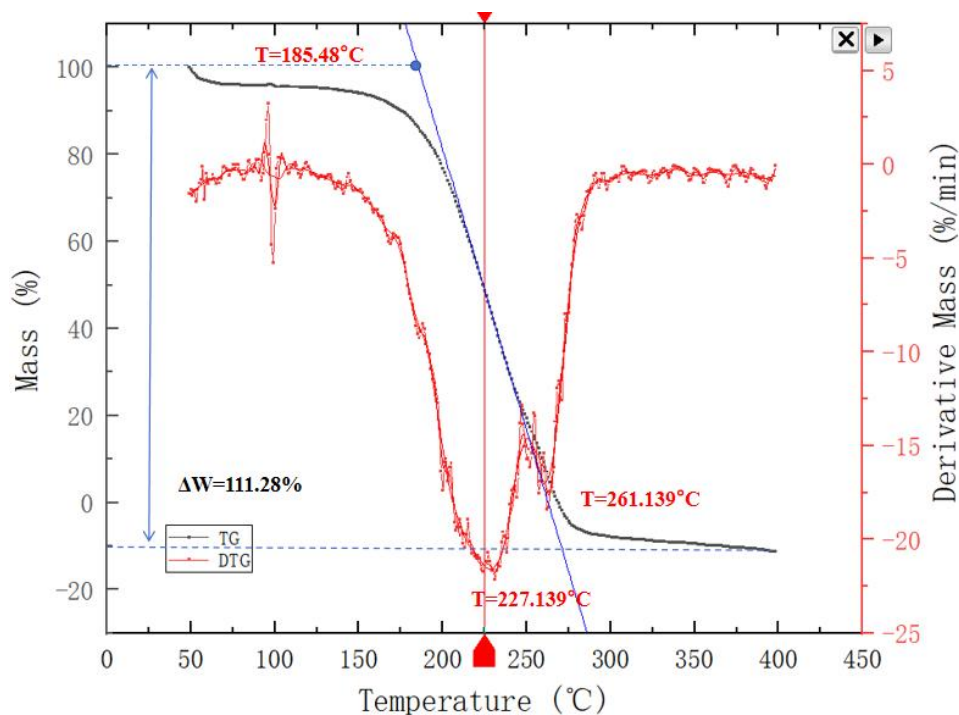


Figure 31. TG and DTG spectrum of stearic acid

According to the thermogravimetric analysis image, the temperature at which sample A1 begins to undergo thermal decomposition reaction is 185.48 ° C, and according to the investigation literature, the temperature at which stearic acid begins to decompose is approximately 223 ° C. The reason for such a large deviation may be that the heating rate is too fast for stearic acid, leading to thermal hysteresis, and the sample size is too small, resulting in a faster rate of reaching thermal equilibrium and an earlier reaction.

After 200 ° C, the quality decrease becomes significant. According to the DTG curve, the first peak appears at 227.139 ° C, while the second peak quickly appears at 261.139 ° C during the same quality decrease stage. The peak of the DTG curve represents the temperature point with the highest rate of mass change, where there are two different thermal events in the same temperature range. This may represent different degrees of C-C bond breakage to form small molecules (Such as methane, ethylene, acetylene, propylene, etc) that reach their respective peak reaction rates.

At the end of the reaction, it was found that the quality had decreased by 111.28%, which was greater than 100%. The reason for this situation may not only be due to errors in the instrument itself, but also because the thermal decomposition products evaporate from the sample and then condense at low temperatures. If these condensates are on the crucible, it will cause the measured weight loss results to be lower. As the temperature further increases, these condensates evaporate again, causing false weight loss and deformation of the TG curve. And the initial reference mass is smaller than the sample mass.

5.1.2 Analyze of DSC and DTG curve

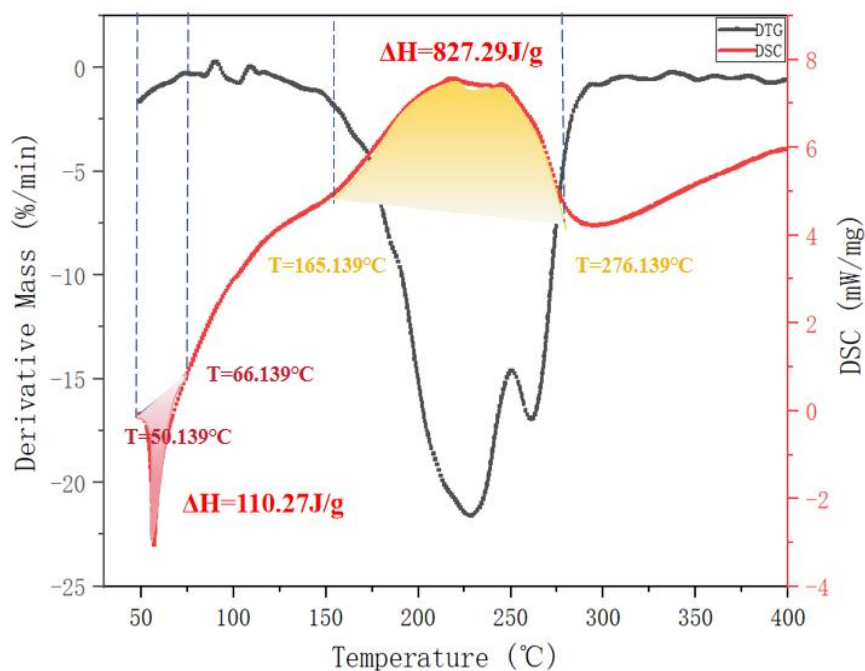


Figure 32.DSC and DTG spectrum of stearic acid

By observing the DSC and DTG curves, it was found that at 50.139 °C, the sample began to absorb heat, while the mass of the sample did not change significantly, possibly due to the evaporation of water present in the sample. The reaction enthalpy of this process is 110.27J/g.

Afterwards, at 165.139 °C, this process underwent a series of complex physical and chemical reactions, such as thermal decomposition, release of small molecule gas products, C-C bond breakage to produce C₂H₄, CH₄, C₂H₂, C₃H₆, and other small molecules, as well as free radicals and long chains broken during the thermal decomposition process. The formed short chain alkanes and alkenes may further recombine to form other products, etc. The reaction enthalpy of this process reached 827.29J/g.

5.2 Sample A2

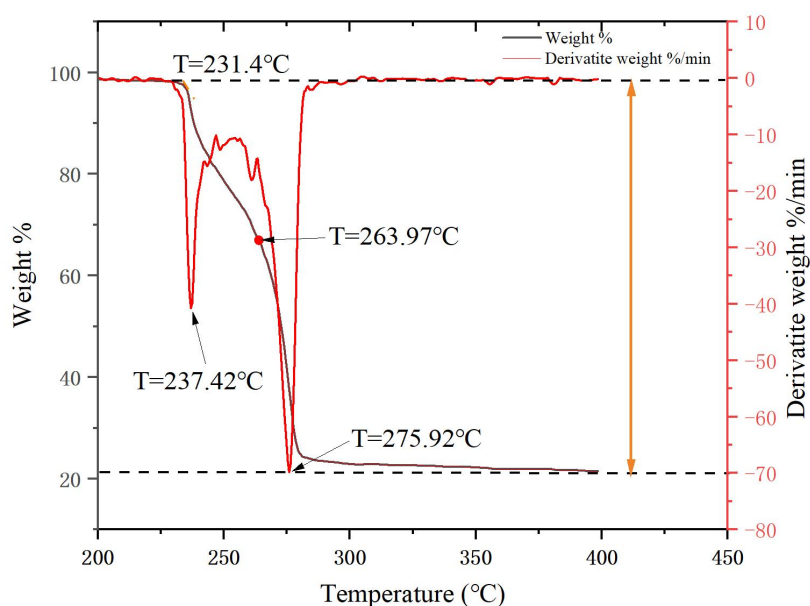
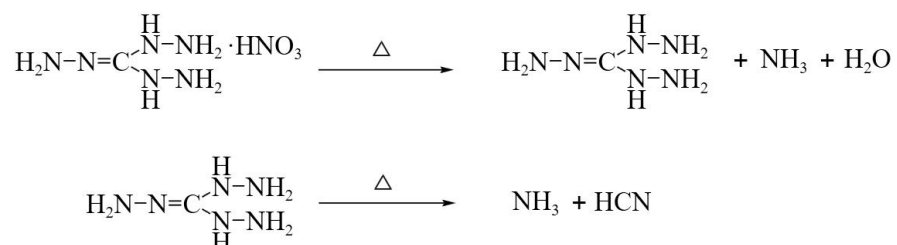


Figure 33. TG figure of sample 2 (black) and DTG figure of sample 2 (red)

As shown in the figure, it can be found that the intersection of the tangent line of the curve with the tangent line of the initial weight of the sample is the starting temperature at which the weight of the sample begins to change as the TG begins to decrease. 220.4°C is close to the boiling point of triamino nitrate (213°C), and the decrease of the mass occurs as a result of evaporation. After reaching the boiling point for some time, it undergoes a slow decrease in mass, reaching the point of maximum decomposition rate at a temperature of (237.42°C). Additionally, at a temperature as high as 263.97 °C, the mass of the sample decreases rapidly and is considered to be near the decomposition temperature (270°C), where the decomposition of the sample is combined with evaporation. The mass decomposition rate peaked at 265.42°C, which is thought to be the decomposition of nitric acid. It stabilized at 300°C, with 21.68% mass remaining and 75.8% mass loss, which is associated with the formation of nitrogen (NH₃), and water (H₂O), hydrogen cyanide (HCN) from the triamino nitrate. The failure of the reaction to reach 100% and the shift in the melting temperature and decomposition temperature compared with the standard ones can be caused by multiple factors such as the experimental environment, the experimental apparatus, and the experimental procedure. The potential reaction is predicted as follows^[13].



5.2.1 Analyze of DTG and TG curves

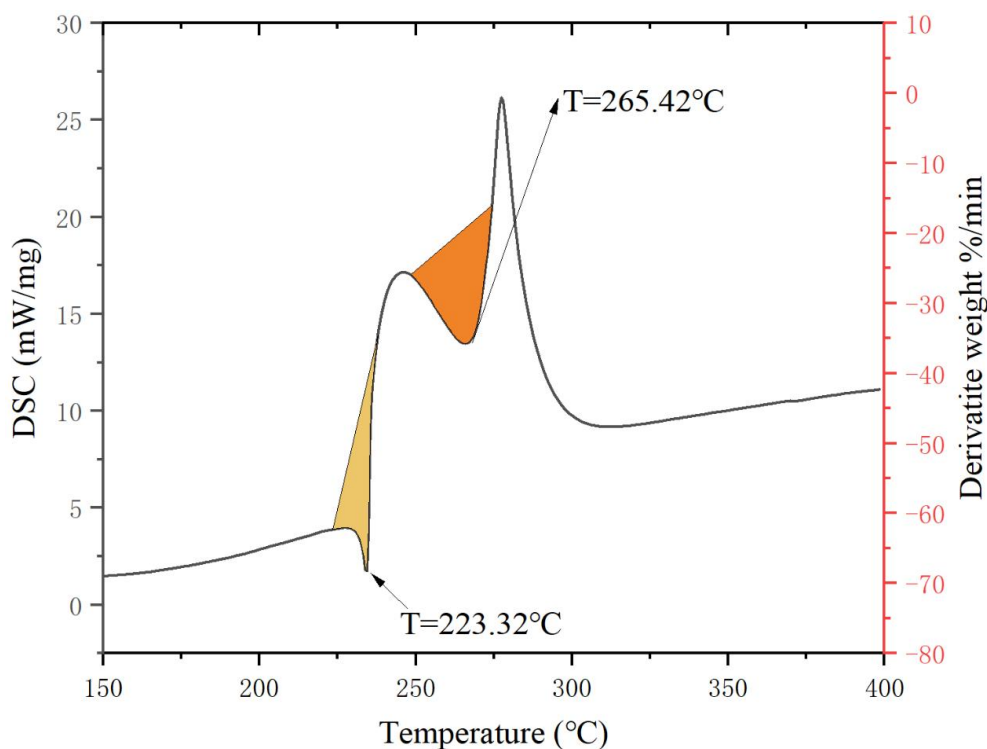


Figure 34. DSC figure of sample 2.

From the DSC image in the above figure, the first peak of DSC should be the heat absorbed by guanidine triamino nitrate in the process of evaporation and removal of nitric acid group during the heating process, and its enthalpy of melting is 136.7 J/g. The second peak of DSC is the heat of reaction absorbed by the decomposition of guanidine triamino nitrate, and its enthalpy of reaction is 340.4 J/g. Through the infrared combined with the thermogravimetric analysis of urea, our group further confirmed that sample 2 is guanidine triamino nitrate.

5.3 Sample A3

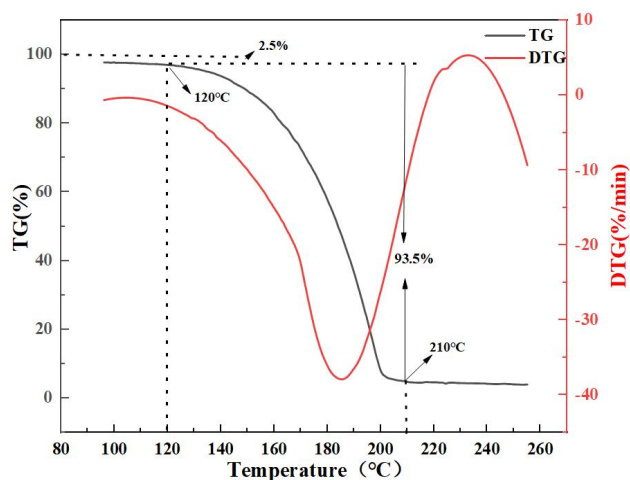


Figure 35. TG figure of product(black) and DTG figure of product(red)

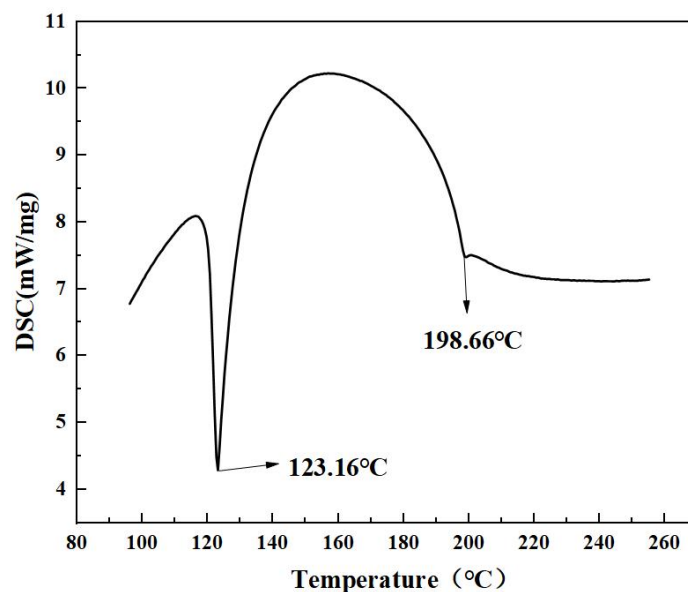


Figure 36. DSC figure of product

As shown in Figure 35 and Figure 36. We can see in TG diagram, mass loss of benzoic acid at 0-120°C is only 2.5%. When the temperature is greater than 120°C, the mass decreases rapidly, and almost all of it is volatilized, only 4.0% of the mass remains after volatilization. For DSC diagram, we can see two distinct endothermic peaks. The starting temperature of the first endothermic peak is 120.21°C, peak temperature is 123.16°C. This endothermic peak is mainly melting peak of benzoic acid from solid to liquid, so its melting point is 123.16°C. The internal molecules of benzoic acid in the crystalline state are recombined under this stability which caused a phase transformation occurs. As a result, the mass loss of benzoic acid at 0-120°C is only 2.5%. The second endothermic peak has a starting temperature of 191.45°C and a peak temperature of 198.66°C. This endothermic peak is mainly the endothermic peak generated by the volatilization of benzoic acid from liquid to gas. The endothermic peak is mainly the endothermic peak generated by the volatilization of benzoic acid from liquid to gas. Hence its boiling point is 198.66°C and benzoic acid will rapidly decrease in mass in the range of 120-200 °C.

5.4 Sample A4

5.4.1 Analyze of DTG and TG curves

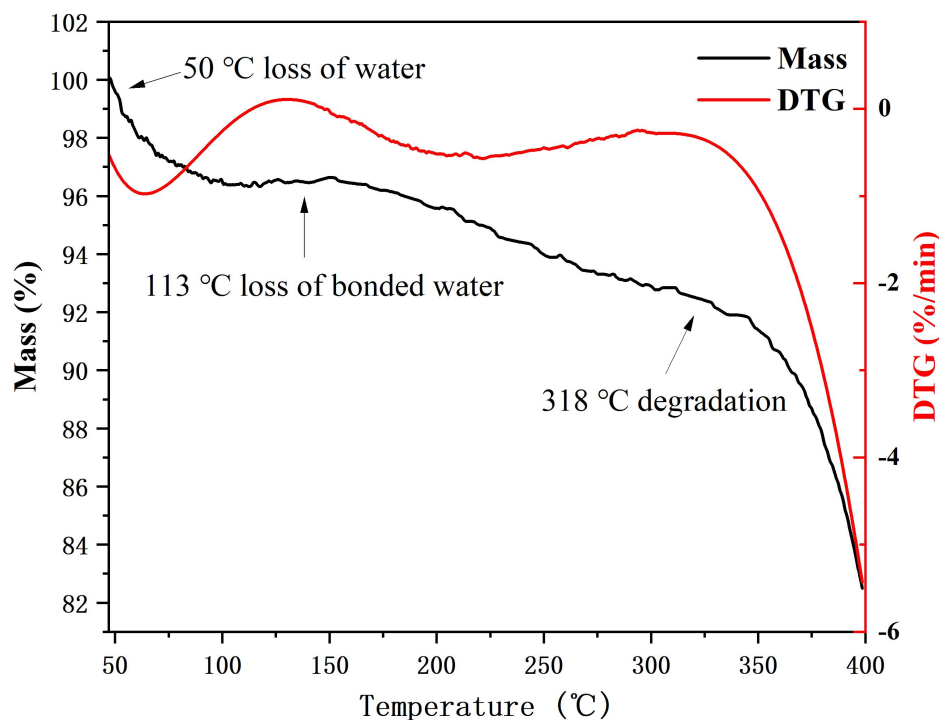


Figure 37. TG and DTG spectrum of Polyvinyl pyrrolidone

From the diagrams there are three temperatures at which there is a clear change in mass, 50°C, 113°C and 318°C. 50°C is the start heating temperature, and the decrease in temperature that occurs from the start of the heating is probably due to sample dampness and therefore evaporation of water during heating. At 113°C, a second phase of mass decrease occurred. From observing the structure of Polyvinyl pyrrolidone, we can see that the nitrogen atom in this substance has a lone pair of electrons, which can easily form a hydrogen bond with water and thus bind to water, the mass loss at this temperature may be due to the separation of water molecules bound to the nitrogen atom. The mass loss at 318°C may be due to the decomposition of Polyvinyl pyrrolidone, and the remaining mass cannot be determined since the mass was decreasing at the end of the heat-up.

5.4.2 Analyze of DSC curves

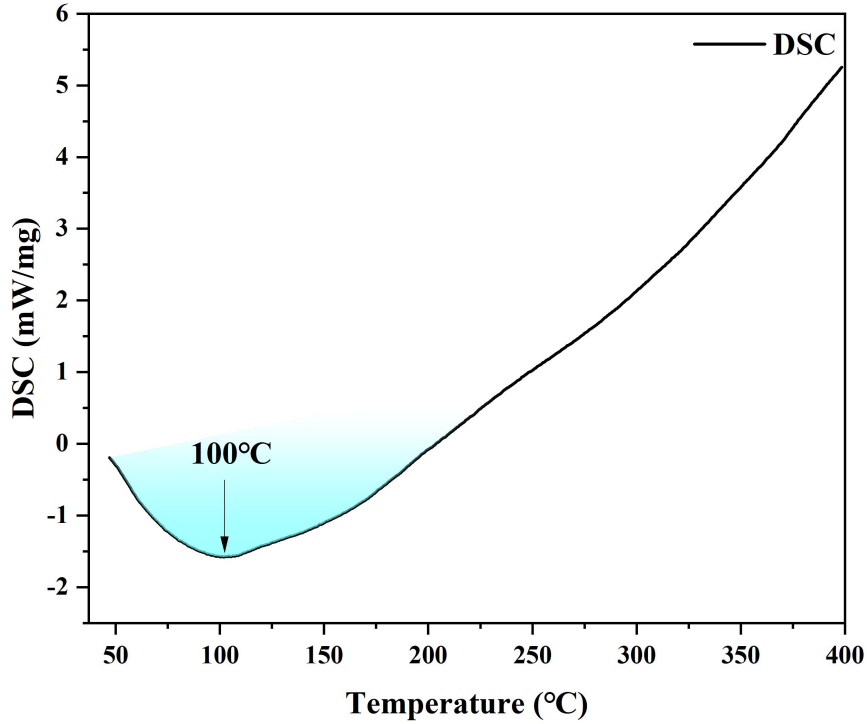


Figure 38. DSC spectrum of Polyvinyl pyrrolidone

In this graph, 100 °C is the lowest point of the peak, and by analyzing the graph, it can be judged that Polyvinyl pyrrolidone absorbs heat at this temperature, and thus this is the crystallization temperature of Polyvinyl pyrrolidone, and by integrating the peaks, the corresponding absorbed heat can be obtained.

5.5 Sample A5

5.5.1 Analysis of TG/DTG curves

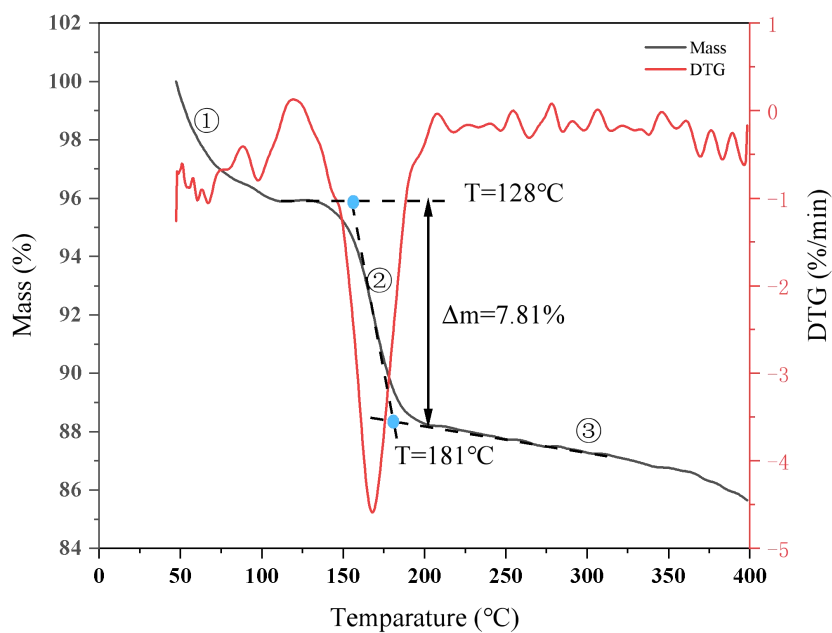


Figure 39 TG (black) and DTG (red) analysis of sample A5

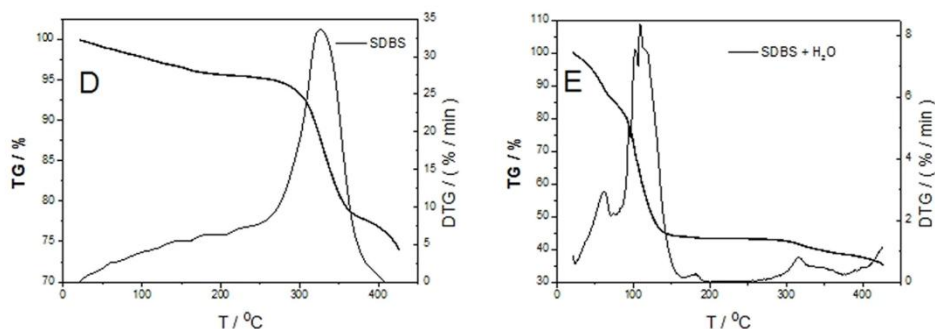


Figure 40. Standard TG and DTG [4] curves of SDBS power (D) and SDBS mixed with water (E)

The DTG curve has a small absorption peak at 90, which shows that the desulfonation reaction has been verified. As can be seen from the figure, regarding the thermochemical efficiency of the STEP degradation reaction of SDBS, the thermochemical effect combined with the desulfonation of SDBS goes through a changing process. It is related to the thermal energy associated with chemical reactions and/or physical transformations such as thermal activation and thermochemical reactions. SDBS experiences a weight loss of less than 5% at the start of stage ① (0 - 181 °C). This might be due to the volatilization of free water. The weight loss reaches 11.81% within the range of 128 - 181 °C (stage ②) and amounts to 14% at around 400 °C. This could be attributed to the vaporization of volatile products, including ionic sulfonates as confirmed by FTIR. However, since the thermal decomposition temperature of SDBS is 450 °C, this experiment is unable to determine the subsequent heating process. Nevertheless, from the process ranging from 0 °C to 400 °C shown in the figure, it can be known that SDBS has been in the weight loss stage all along.

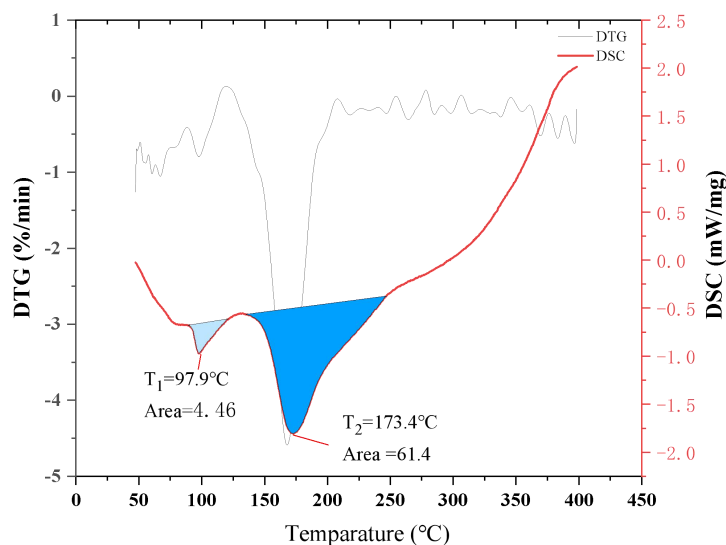


Figure 41. DSC analysis of sample A5

5.5.2 Analysis of DSC curves

For DSC analysis, there are two endothermic peaks. Through analyzing this graph, it can be determined that

SDBS absorbs heat at this temperature. Based on the DTG curve shown above, the absorption peak at T1 is mainly caused by the desulfonation reaction or the volatilization of water/ionic sulfate and so on. And the T2 peak should correspond to the absorbed heat obtained. However, as the thermal decomposition temperature of SDBS is approximately 450 degrees, this experiment should expand the temperature range to further investigate the decomposition condition of SDBS.

6. Conclusion

In this experiment, we completed infrared spectrum testing and thermogravimetric analysis of six samples from group A, and confirmed the composition of each sample through comparison with standard samples and information collection. In the process of writing the report, we have a deep understanding of the experimental principles of infrared spectrum analysis technology and thermogravimetric technology, the main points of instrument use, data processing and conclusion analysis.

For infrared spectrum, we deeply understand the influence of each effect on the peak displacement and the cause of the deviation between the experimental peak and the standard spectrum. For thermogravimetric analysis, We have a deep understanding of the change of the corresponding physical and chemical properties of the sample by controlling the change of temperature and laid a solid foundation for the future research and study. And the final determination perfectly matches with the part 1 which is shown in Table 8.

Table 1. The determination of the material

<i>Sample</i>	<i>Material</i>
A1	Stearic acid
A2	TAGN
A3	Benzoic acid
A4	Polyvinyl pyrrolidone
A5	SDBS
A6	Dopamine

7. Contributions

Group members	Task Assignment
Liu Yuxin	1. Manage and organize the group members as group leader 1. Complete the analysis of the infrared spectropy sample A-6. 3. Write the determination of Material and Conclusion 4. Write the part of introduction of TG.
Wang Zhuoyu	1. Complete the analysis of the infrared spectropy sample A-4. 2. Write and graph the analysis of sample A-4 in TG DTG and DSC 3. Write the part of introduction of ATR and FTIR.
Zhao Bokun	1. Complete the infrared spectrum analysis and calibration of sample A-2. 2. Write and graph the analysis of sample A-1 in TG DTG and DSC 3. Provide an introduction of IR Probing machines and experimental process..
Yang Chengrui	1. Complete the analysis of the infrared spectropy sample A-2. 2. Write and graph the analysis of sample A-2 in TG DTG and DSC 3. Write the part of Definition of infrared ray , introduction to IR ,and the theories of IR
Jin Zeyin	1. Complete the infrared spectrum analysis and calibration of sample A-2. 2. Write and graph the analysis of sample A-1 in TG DTG and DSC 3. Study of the problematic approach to the processing of version 1.0 data at OMNIC
Hou Zherui	1. Complete the infrared spectrum analysis and calibration of sample A-2. 2. Write and graph the analysis of sample A-1 in TG DTG and DSC 3. Provide an introduction to the DSC technology.

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